

DUNEDIN, New Zealand

WMO: 938910

Lat: 45.9267S

Lon: 170.1967E

Elev: 7

StdP: 14.69

Time Zone: 12.00 (NZL)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 24.3	(c) 26.6	(d) 23.3	(e) 17.6	(f) 26.5	(g) 25.6	(h) 19.6	(i) 28.6	(j) 27.0	(k) 48.4	(l) 23.9	(m) 47.8	(n) 1.7	(o) 200	(p) 0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 18.8	(c) 79.9	(d) 63.5	(e) 76.2	(f) 62.2	(g) 73.0	(h) 60.9	(i) 65.6	(j) 76.1	(k) 63.8	(l) 73.4	(m) 62.2	(n) 70.8	(o) 11.1	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.6	82.0	69.6	59.8	76.9	67.5	58.3	72.7	65.8	30.4	76.8	29.1	73.3	28.0	70.9	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.3	21.9	19.1		DB	21.2	87.2	2.2	2.9	19.7	89.3	18.4	91.0	17.2	92.6	15.6	94.8
				WB	20.8	69.5	2.2	1.7	19.2	70.7	17.8	71.7	16.6	72.7	15.0	73.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	59.4	58.5	56.1	51.2	46.6	42.1	41.4	43.9	47.5	50.1	53.9	57.4
	DBStd	8.06	5.39	5.36	5.58	5.20	5.10	4.95	5.07	4.62	4.98	4.99	5.55	5.52
	HDD50	1085	1	1	8	45	129	241	269	197	107	60	22	5
	HDD65	5288	190	192	282	413	570	688	730	655	526	464	335	243
	CDD50	1315	291	240	197	82	25	4	4	7	30	62	138	235
	CDD65	43	16	11	6	0	0	0	0	0	0	1	1	8
	CDH74	569	219	129	67	3	0	0	0	0	0	10	38	103
	CDH80	104	50	24	10	0	0	0	0	0	0	1	4	15
Wind	WSAvg	7.4	8.1	7.4	6.8	6.6	6.4	6.7	6.3	6.3	8.0	8.6	8.8	8.3
Precipitation	PrecAvg	27.5	2.8	2.2	2.2	2.2	2.8	2.2	2.0	2.0	1.8	2.4	2.3	2.7
	PrecMax	37.0	7.3	4.3	6.3	6.3	9.2	6.3	5.1	5.1	4.1	5.4	4.8	4.3
	PrecMin	20.8	0.8	0.7	0.5	0.4	0.5	0.5	0.6	0.6	0.6	0.5	0.7	0.7
	PrecStd	4.0	1.6	1.1	1.1	1.2	2.0	1.5	1.1	1.0	0.7	1.1	1.0	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.3	83.7	80.6	73.1	67.6	61.8	60.2	63.4	70.3	74.6	78.9	82.1
		MCWB	66.1	65.1	63.6	58.1	55.5	51.3	49.4	51.0	54.9	58.1	62.4	63.9
	2%	DB	79.6	78.1	74.9	68.4	62.4	57.1	56.5	59.0	65.1	68.6	73.4	76.8
		MCWB	63.4	63.7	61.7	56.2	52.3	49.3	47.9	49.6	52.9	55.7	59.4	62.9
	5%	DB	75.4	73.4	70.9	64.7	59.2	53.7	53.5	55.9	61.5	64.5	69.5	72.5
		MCWB	62.2	62.1	60.5	55.5	51.8	47.9	46.6	48.5	51.3	53.6	57.9	60.8
	10%	DB	71.1	70.0	67.4	61.5	56.3	50.9	50.5	53.0	58.4	61.2	65.6	68.8
		MCWB	60.6	60.7	58.7	54.2	50.1	46.4	45.4	47.2	49.9	52.4	56.1	59.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.2	68.5	66.3	60.3	57.2	52.7	50.9	52.4	56.7	60.3	64.1	66.2
		MCDB	79.0	77.6	76.4	69.1	65.6	59.2	57.0	60.3	66.7	70.9	75.6	77.4
	2%	WB	65.4	65.1	63.4	58.1	54.1	50.1	49.0	50.6	54.1	57.3	61.1	63.9
		MCDB	76.3	74.3	72.0	65.0	60.2	55.3	54.5	57.0	62.6	66.8	71.1	74.5
	5%	WB	63.6	63.1	61.2	56.4	52.2	48.4	47.4	49.1	52.2	55.0	58.7	61.9
		MCDB	72.9	72.1	69.2	62.6	57.4	52.5	52.0	54.5	59.8	62.8	67.3	70.4
	10%	WB	61.5	61.1	59.3	54.7	50.8	46.9	45.8	47.7	50.5	52.9	56.6	60.0
		MCDB	69.6	68.8	66.3	60.3	55.2	50.7	50.0	52.1	57.1	59.8	64.1	67.4
Mean Daily Temperature Range		MDBR	18.8	18.9	19.4	18.5	16.8	15.9	16.9	17.5	18.3	18.3	18.1	17.6
	5% DB	MCDBR	27.9	26.0	26.3	24.6	22.5	20.2	20.9	22.9	24.6	25.6	25.7	24.8
		MCWBR	14.2	14.0	14.6	14.9	14.8	13.8	14.0	15.4	15.0	15.1	13.9	12.8
	5% WB	MCDBR	24.5	23.0	23.4	21.3	19.8	18.3	18.7	20.2	22.2	23.2	23.5	22.4
MCWBR		13.6	13.0	14.1	13.8	13.9	13.4	13.3	14.3	14.4	14.5	13.5	12.4	
Clear-Sky Solar Irradiance	taub	0.338	0.332	0.312	0.304	0.290	0.288	0.281	0.305	0.319	0.322	0.330	0.333	
	taud	2.525	2.554	2.617	2.612	2.613	2.575	2.604	2.561	2.509	2.522	2.514	2.543	
	Ebn at Noon	311	304	295	273	252	237	252	268	287	304	312	315	
	Edh at Noon	34	31	26	22	18	16	17	23	29	32	34	34	
All-Sky Solar Radiation	RadAvg	1859	1641	1216	780	470	357	436	667	1059	1490	1818	1963	
	RadStd	142	128	55	52	29	27	28	41	61	84	92	121	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	+1.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page